Exploratory Analysis of Global Earthquake-Tsunami Events (2001–2022)

Importing the libraries

In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the data

In [52]:
```
df = pd.read_csv("earthquake_data_tsunami.csv")
print("Dataset loaded successfully!\n")
print("Shape of the dataset:", df.shape)
print("\nColumn Names:\n", df.columns.tolist())

df.describe()
```

```
Dataset loaded successfully!

Shape of the dataset: (782, 13)

Column Names:
 ['magnitude', 'cdi', 'mmi', 'sig', 'nst', 'dmin', 'gap', 'depth', 'latitude', 'longitude', 'Year', 'Month', 'tsunami']
```

Out[52]:

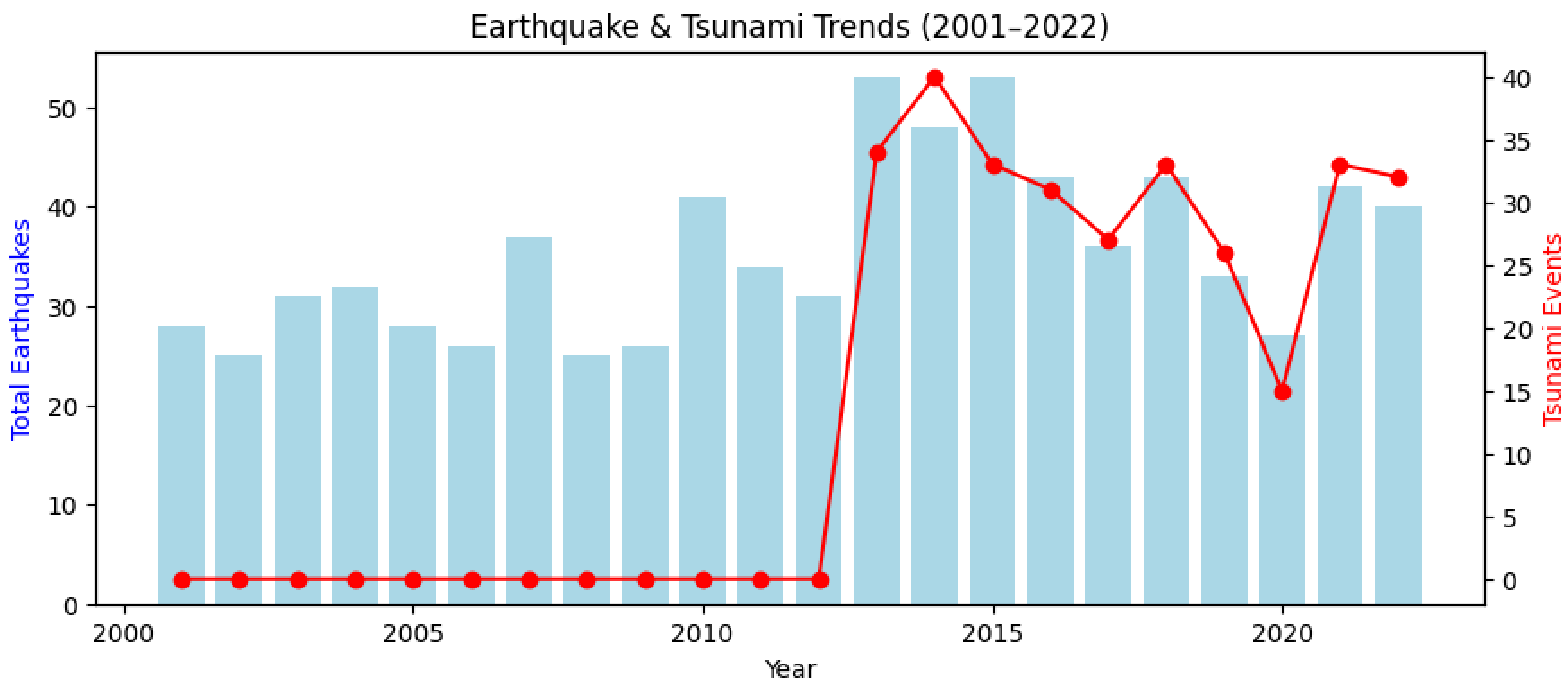
	magnitude	cdi	mmi	sig	nst	dmin	gap	depth	latitude	longitude	'
count	782.000000	782.000000	782.000000	782.000000	782.000000	782.000000	782.000000	782.000000	782.000000	782.000000	782.00(
mean	6.941125	4.333760	5.964194	870.108696	230.250639	1.325757	25.038990	75.883199	3.538100	52.609199	2012.28(
std	0.445514	3.169939	1.462724	322.465367	250.188177	2.218805	24.225067	137.277078	27.303429	117.898886	6.09(
min	6.500000	0.000000	1.000000	650.000000	0.000000	0.000000	0.000000	2.700000	-61.848400	-179.968000	2001.00(
25%	6.600000	0.000000	5.000000	691.000000	0.000000	0.000000	14.625000	14.000000	-14.595600	-71.668050	2007.00(
50%	6.800000	5.000000	6.000000	754.000000	140.000000	0.000000	20.000000	26.295000	-2.572500	109.426000	2013.00(
75%	7.100000	7.000000	7.000000	909.750000	445.000000	1.863000	30.000000	49.750000	24.654500	148.941000	2017.00(
max	9.100000	9.000000	9.000000	2910.000000	934.000000	17.654000	239.000000	670.810000	71.631200	179.662000	2022.00(

1. Time-Based Analysis:

o Explore how earthquake occurrences and tsunami events have changed over the 22-year period (2001–2022).

In [54]:

```
df['Year'] = pd.to_numeric(df['Year'], errors='coerce')
df['tsunami'] = pd.to_numeric(df['tsunami'], errors='coerce')

data = df[(df['Year'] >= 2001) & (df['Year'] <= 2022)]
yearly = data.groupby('Year').agg({'tsunami':'sum','magnitude':'count'}).rename(columns={'magnitude':'Total_Earthquakes','tsun

fig, ax1 = plt.subplots(figsize=(10,4))
ax1.bar(yearly.index, yearly['Total_Earthquakes'], color='lightblue', label='Total Earthquakes')
ax2 = ax1.twinx()
ax2.plot(yearly.index, yearly['Tsunami_Events'], color='red', marker='o', label='Tsunami Events')

ax1.set_xlabel('Year'); ax1.set_ylabel('Total Earthquakes', color='blue')
ax2.set_ylabel('Tsunami Events', color='red')
plt.title('Earthquake & Tsunami Trends (2001–2022)')
plt.show()
```

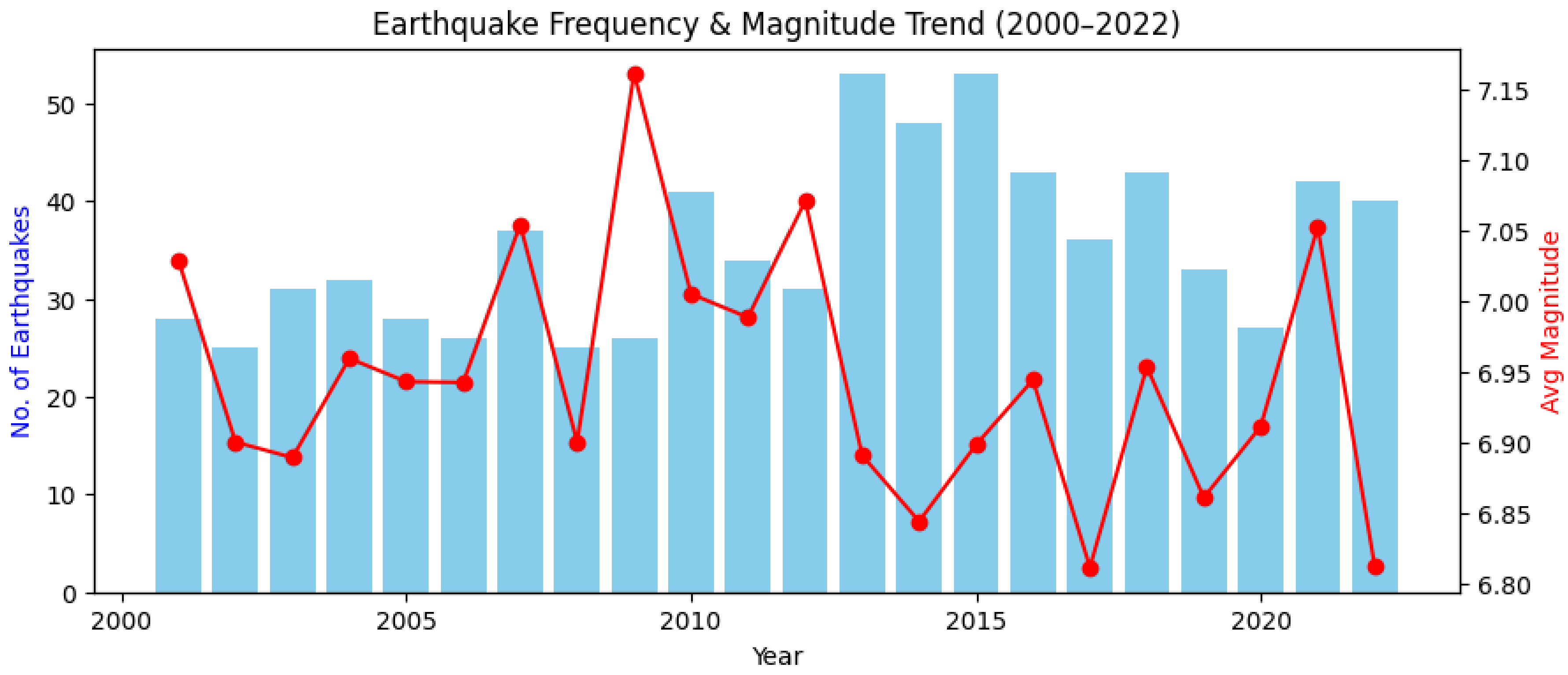


o Identify any trends in the frequency or magnitude of earthquakes over time.

In [53]:

```
df['Year'] = pd.to_numeric(df['Year'], errors='coerce')
df['magnitude'] = pd.to_numeric(df['magnitude'], errors='coerce')

data = df.groupby('Year')['magnitude'].agg(['count', 'mean']).loc[2000:2022]

fig, ax1 = plt.subplots(figsize=(10,4))
ax1.bar(data.index, data['count'], color='skyblue', label='Total Earthquakes')
ax2 = ax1.twinx()
ax2.plot(data.index, data['mean'], color='red', marker='o', label='Avg Magnitude')

ax1.set_xlabel('Year'); ax1.set_ylabel('No. of Earthquakes', color='blue')
ax2.set_ylabel('Avg Magnitude', color='red')
```

```
plt.title('Earthquake Frequency & Magnitude Trend (2000-2022)')
plt.show()
```

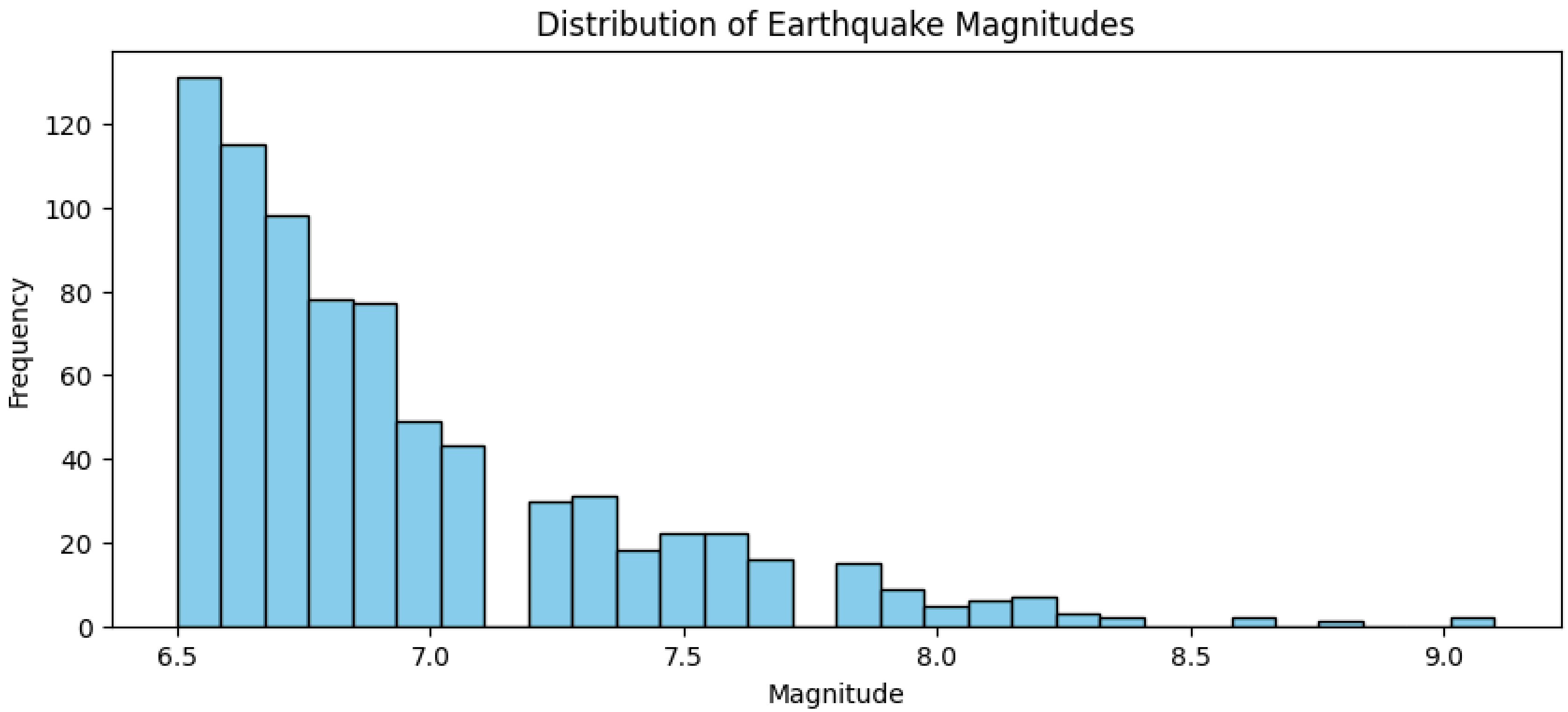


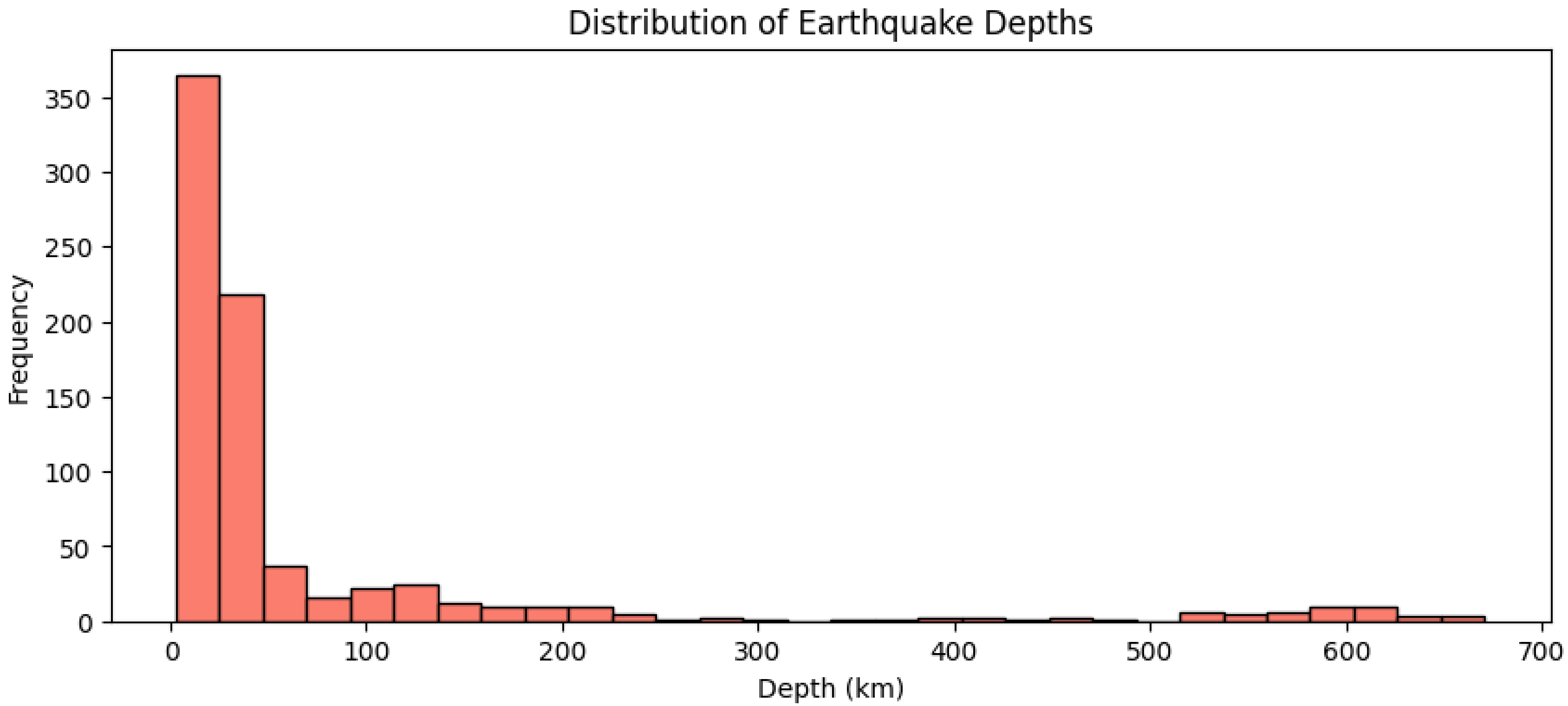
2. Magnitude and Depth Analysis:

o Analyze the distribution of earthquake magnitudes and depths.

In [11]:
```
df['magnitude'] = pd.to_numeric(df['magnitude'], errors='coerce')
df['depth'] = pd.to_numeric(df['depth'], errors='coerce')

# --- Plot Magnitude Distribution ---
plt.figure(figsize=(10,4))
plt.hist(df['magnitude'].dropna(), bins=30, color='skyblue', edgecolor='black')
plt.title('Distribution of Earthquake Magnitudes')
plt.xlabel('Magnitude')
```

```
plt.ylabel('Frequency')
plt.show()

# --- Plot Depth Distribution ---
plt.figure(figsize=(10,4))
plt.hist(df['depth'].dropna(), bins=30, color='salmon', edgecolor='black')
plt.title('Distribution of Earthquake Depths')
plt.xlabel('Depth (km)')
plt.ylabel('Frequency')
plt.show()
```



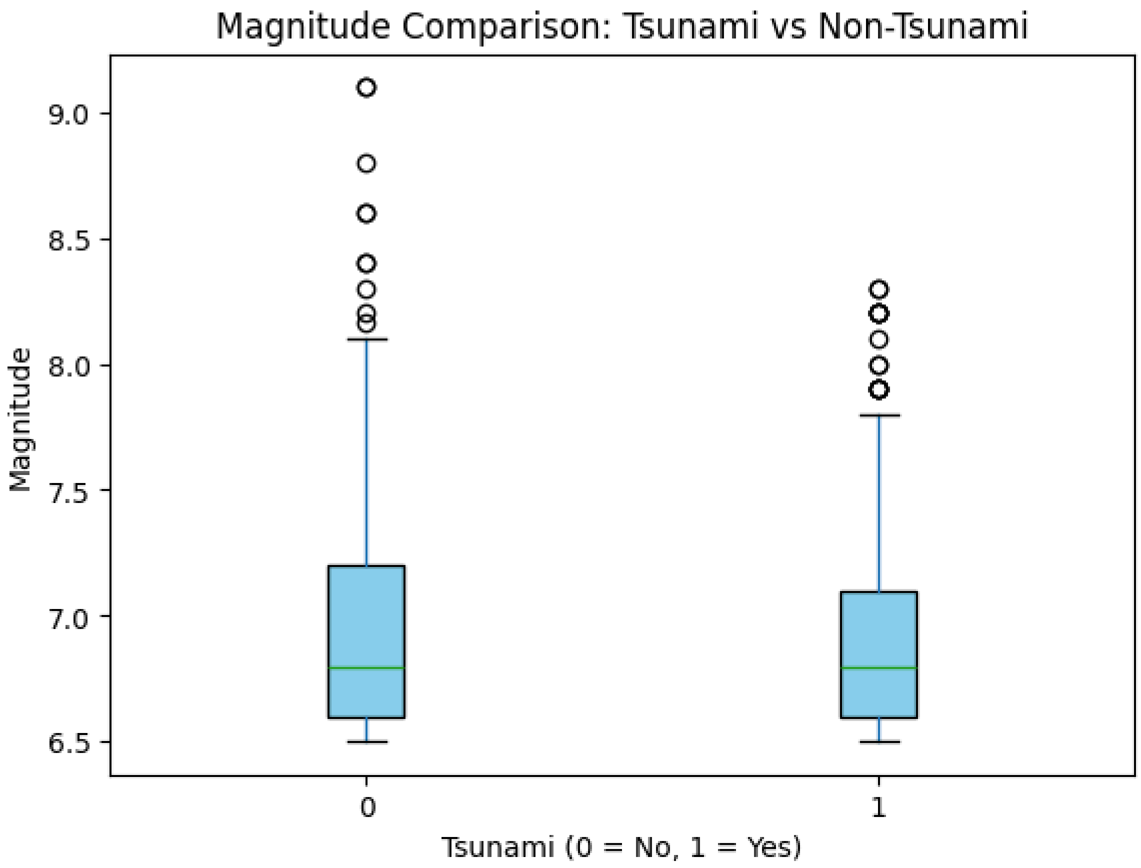


o Compare the average magnitude and depth of tsunami vs. non-tsunami events.

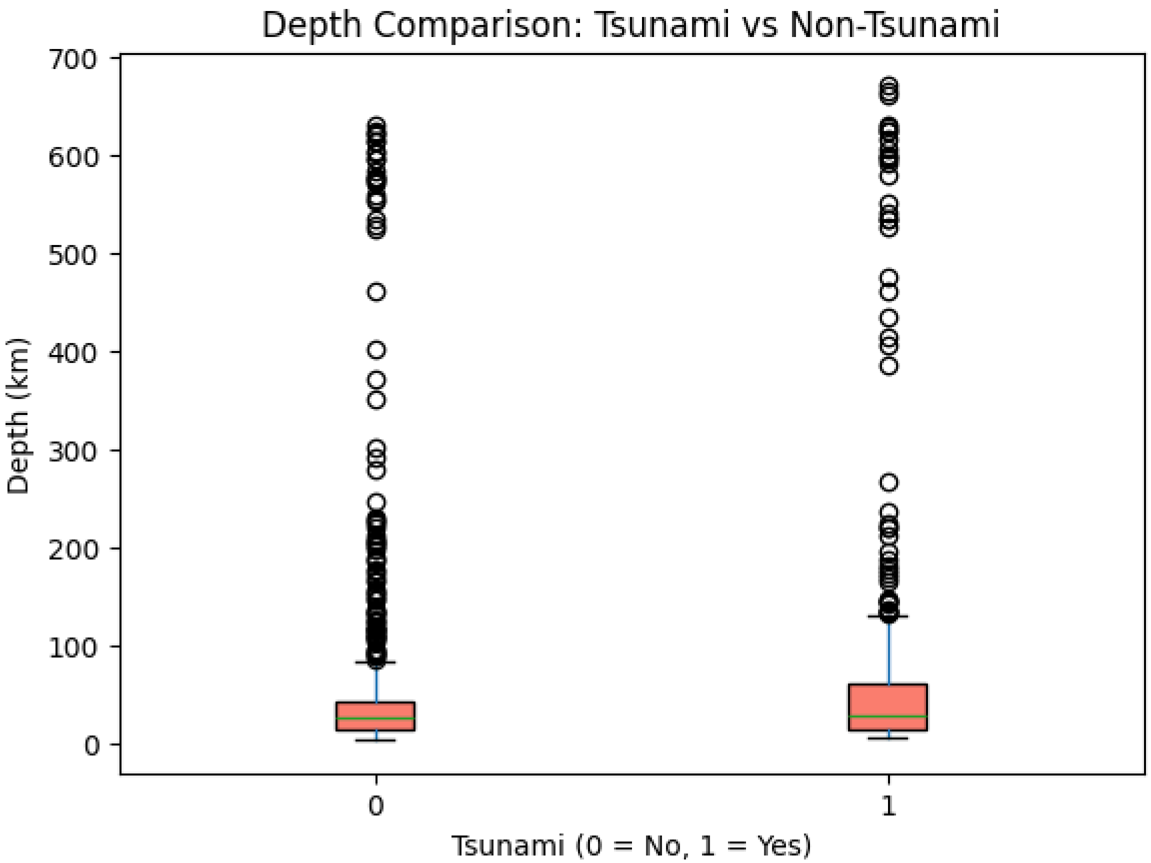
In [17]:
```
df['magnitude'] = pd.to_numeric(df['magnitude'], errors='coerce')
df['depth'] = pd.to_numeric(df['depth'], errors='coerce')
df['tsunami'] = pd.to_numeric(df['tsunami'], errors='coerce')

# --- Boxplot for Magnitude ---
plt.figure(figsize=(8,5))
df.boxplot(column='magnitude', by='tsunami', grid=False,
           patch_artist=True, boxprops=dict(facecolor='skyblue'))
plt.title('Magnitude Comparison: Tsunami vs Non-Tsunami')
plt.suptitle('')
plt.xlabel('Tsunami (0 = No, 1 = Yes)')
plt.ylabel('Magnitude')
plt.show()
```

```python
# --- Boxplot for Depth ---
plt.figure(figsize=(8,5))
df.boxplot(column='depth', by='tsunami', grid=False,
           patch_artist=True, boxprops=dict(facecolor='salmon'))
plt.title('Depth Comparison: Tsunami vs Non-Tsunami')
plt.suptitle('')
plt.xlabel('Tsunami (0 = No, 1 = Yes)')
plt.ylabel('Depth (km)')
plt.show()
```

```
<Figure size 800x500 with 0 Axes>
```




```
<Figure size 800x500 with 0 Axes>
```



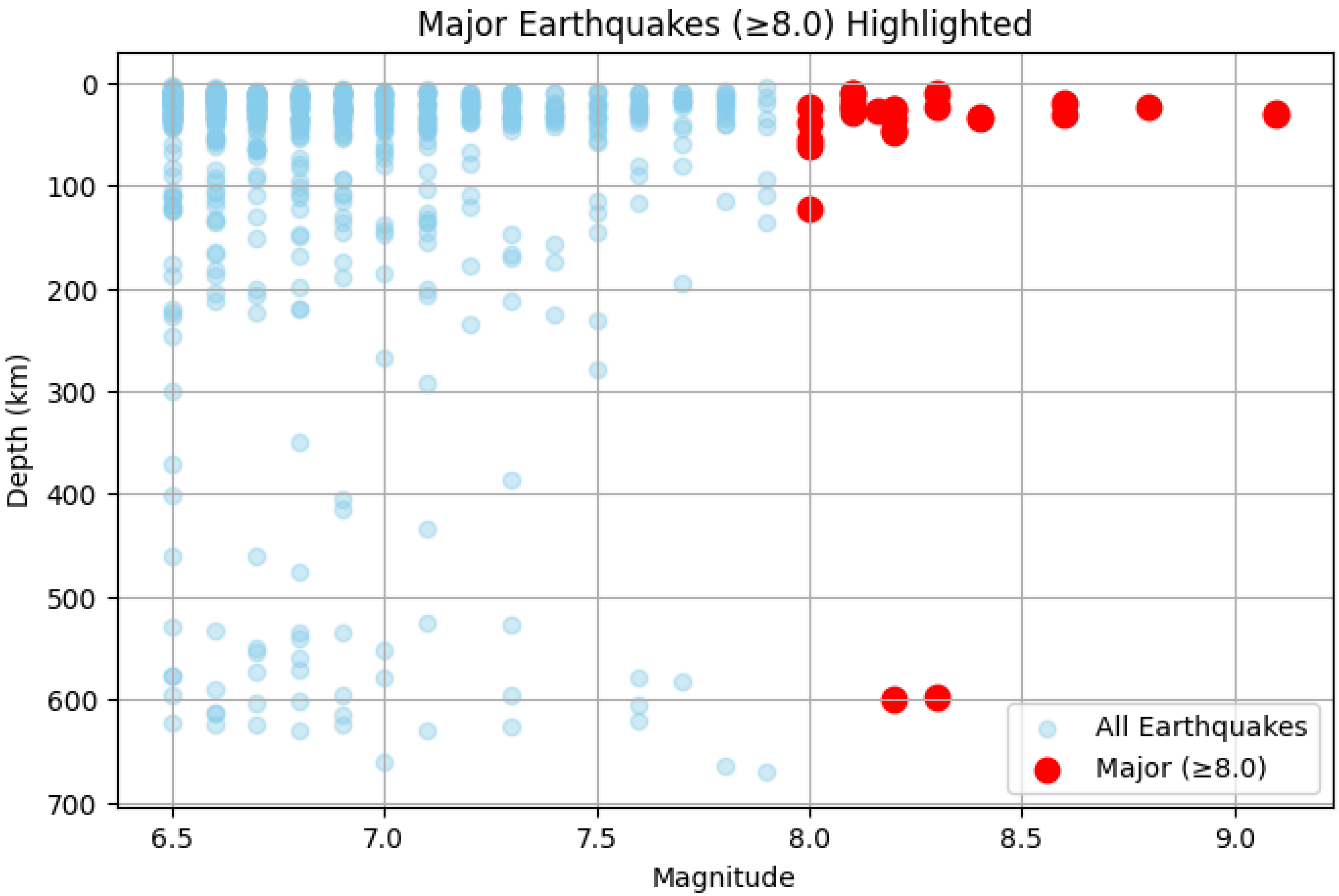
o **Highlight major earthquakes (≥8.0) and their characteristics.**

In [21]:
```
df['magnitude'] = pd.to_numeric(df['magnitude'], errors='coerce')
df['depth'] = pd.to_numeric(df['depth'], errors='coerce')
df['tsunami'] = pd.to_numeric(df['tsunami'], errors='coerce')

# Filter major earthquakes (≥ 8.0)
major_eq = df[df['magnitude'] >= 8.0]

# --- Display key characteristics ---
print("✅ Major Earthquakes (Magnitude ≥ 8.0):")
display(major_eq[['magnitude', 'depth', 'latitude', 'longitude', 'Year', 'Month', 'tsunami']].head(15))
```

```python
# --- Scatter Plot (Magnitude vs Depth) ---
plt.figure(figsize=(8,5))
plt.scatter(df['magnitude'], df['depth'], alpha=0.4, color='skyblue', label='All Earthquakes')
plt.scatter(major_eq['magnitude'], major_eq['depth'], color='red', s=80, label='Major (≥8.0)')
plt.gca().invert_yaxis()  # Depth increases downward
plt.title('Major Earthquakes (≥8.0) Highlighted')
plt.xlabel('Magnitude')
plt.ylabel('Depth (km)')
plt.legend()
plt.grid(True)
plt.show()
```

✅ Major Earthquakes (Magnitude ≥ 8.0):

	magnitude	depth	latitude	longitude	Year	Month	tsunami
56	8.1	22.79	-58.4157	-25.3206	2021	8	0
59	8.2	46.66	55.4742	-157.9170	2021	7	1
60	8.2	35.00	55.3154	-157.8290	2021	7	1
74	8.1	28.93	-29.7466	-177.2240	2021	3	1
129	8.0	122.57	-5.8119	-75.2697	2019	5	1
170	8.2	600.00	-18.1125	-178.1530	2018	8	1
198	8.2	47.39	15.0222	-93.8993	2017	9	1
285	8.3	22.44	-31.5729	-71.6744	2015	9	1
356	8.2	25.00	-19.6097	-70.7691	2014	4	1
393	8.3	598.10	54.8920	153.2210	2013	5	1
414	8.0	24.00	-10.7990	165.1140	2013	2	1
440	8.2	25.10	0.8020	92.4630	2012	4	0
441	8.6	20.00	2.3270	93.0630	2012	4	0
476	9.1	29.00	38.2970	142.3730	2011	3	0
517	8.8	22.90	-36.1220	-72.8980	2010	2	0

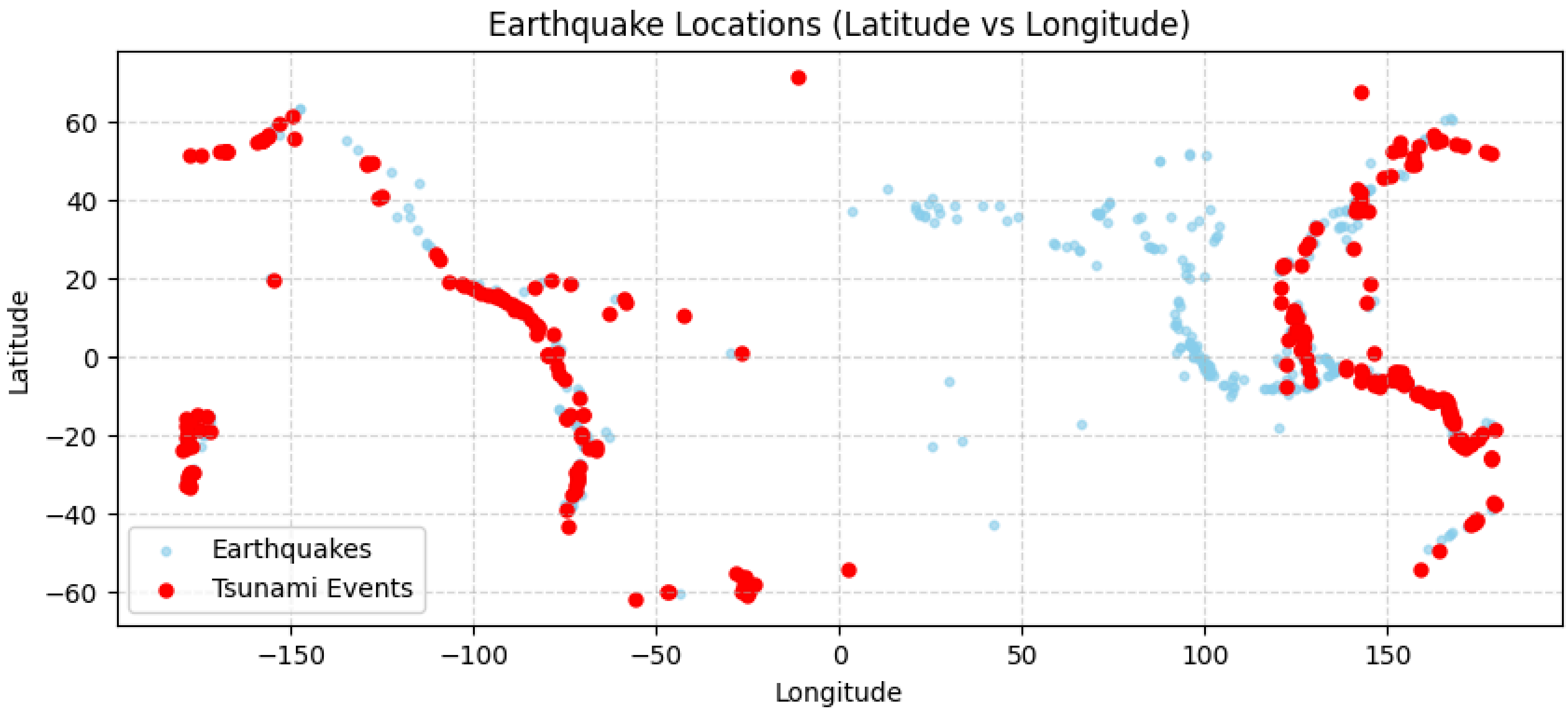


3. Geographic Distribution Using 2D Plotting:

o Plot earthquake locations using latitude and longitude on a 2D scatter plot.

In [55]:
```
df['latitude'] = pd.to_numeric(df['latitude'], errors='coerce')
df['longitude'] = pd.to_numeric(df['longitude'], errors='coerce')
df['tsunami'] = pd.to_numeric(df['tsunami'], errors='coerce')

# --- Plot Earthquake Locations ---
plt.figure(figsize=(10,4))
```

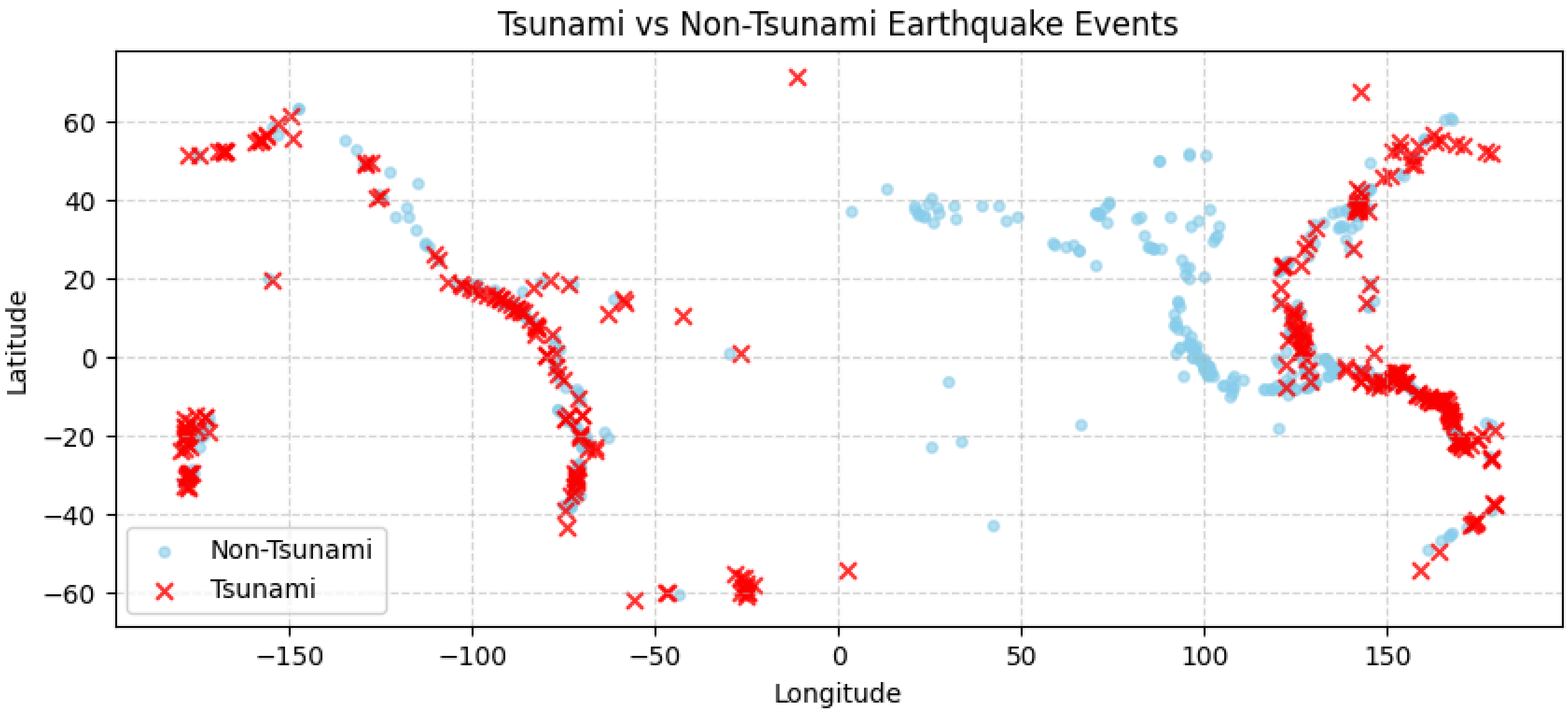
```
plt.scatter(df['longitude'], df['latitude'], s=10, alpha=0.6, color='skyblue', label='Earthquakes')
plt.scatter(df[df['tsunami']==1]['longitude'], df[df['tsunami']==1]['latitude'],
            s=25, color='red', label='Tsunami Events')

plt.title('Earthquake Locations (Latitude vs Longitude)')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.legend()
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```



o Visually distinguish between tsunami and non-tsunami events.

In [56]:
```
df['latitude'] = pd.to_numeric(df['latitude'], errors='coerce')
df['longitude'] = pd.to_numeric(df['longitude'], errors='coerce')
df['tsunami'] = pd.to_numeric(df['tsunami'], errors='coerce')
```

```
# --- Scatter Plot ---
plt.figure(figsize=(10,4))
plt.scatter(df[df['tsunami']==0]['longitude'], df[df['tsunami']==0]['latitude'],
            s=15, color='skyblue', alpha=0.6, label='Non-Tsunami')
plt.scatter(df[df['tsunami']==1]['longitude'], df[df['tsunami']==1]['latitude'],
            s=40, color='red', alpha=0.8, marker='x', label='Tsunami')

plt.title('Tsunami vs Non-Tsunami Earthquake Events')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.legend()
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```



o Identify clusters or regions with higher concentration of tsunami events (without using map tiles or interactive maps).

In [57]:
```
df['latitude'] = pd.to_numeric(df['latitude'], errors='coerce')
df['longitude'] = pd.to_numeric(df['longitude'], errors='coerce')
df['tsunami'] = pd.to_numeric(df['tsunami'], errors='coerce')

tsunami_df = df[df['tsunami'] == 1].copy()
tsunami_df['lat_grp'] = tsunami_df['latitude'].round(1)
tsunami_df['lon_grp'] = tsunami_df['longitude'].round(1)

# Count tsunami events per region
region = (tsunami_df.groupby(['lat_grp','lon_grp'])
          .size().reset_index(name='Tsunami_Events')
          .sort_values('Tsunami_Events', ascending=False))

# Display top regions and summary
print(" Top Regions with Highest Tsunami Concentration:\n")
display(region.head(10))

total = len(tsunami_df)
share = region.head(10)['Tsunami_Events'].sum() / total * 100
print(f"Total Tsunami Events: {total}")
print(f"Top 10 regions cover about {share:.2f}% of all tsunami events.")
```

```
Top Regions with Highest Tsunami Concentration:
```

	lat_grp	lon_grp	Tsunami_Events
116	-11.0	165.7	2
48	-26.0	178.4	2
273	52.5	-168.1	2
274	52.5	-167.7	2
136	-6.8	155.0	2
218	14.7	-92.5	2
1	-60.9	-25.1	1
2	-60.3	-47.1	1
3	-60.3	-46.4	1
4	-60.3	-25.0	1

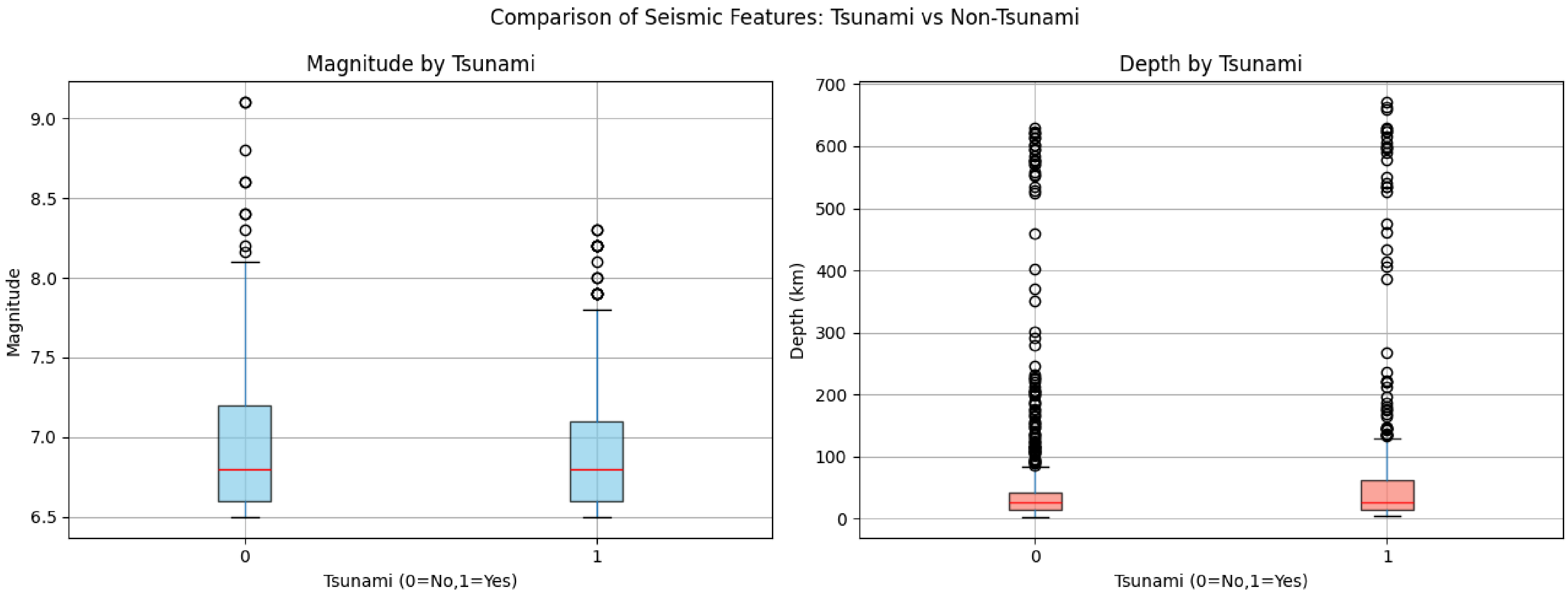
```
Total Tsunami Events: 304
Top 10 regions cover about 5.26% of all tsunami events.
```

4. Statistical and Comparative Analysis:

o Use box plots, histograms, and bar charts to compare seismic features between tsunami and non-tsunami events.

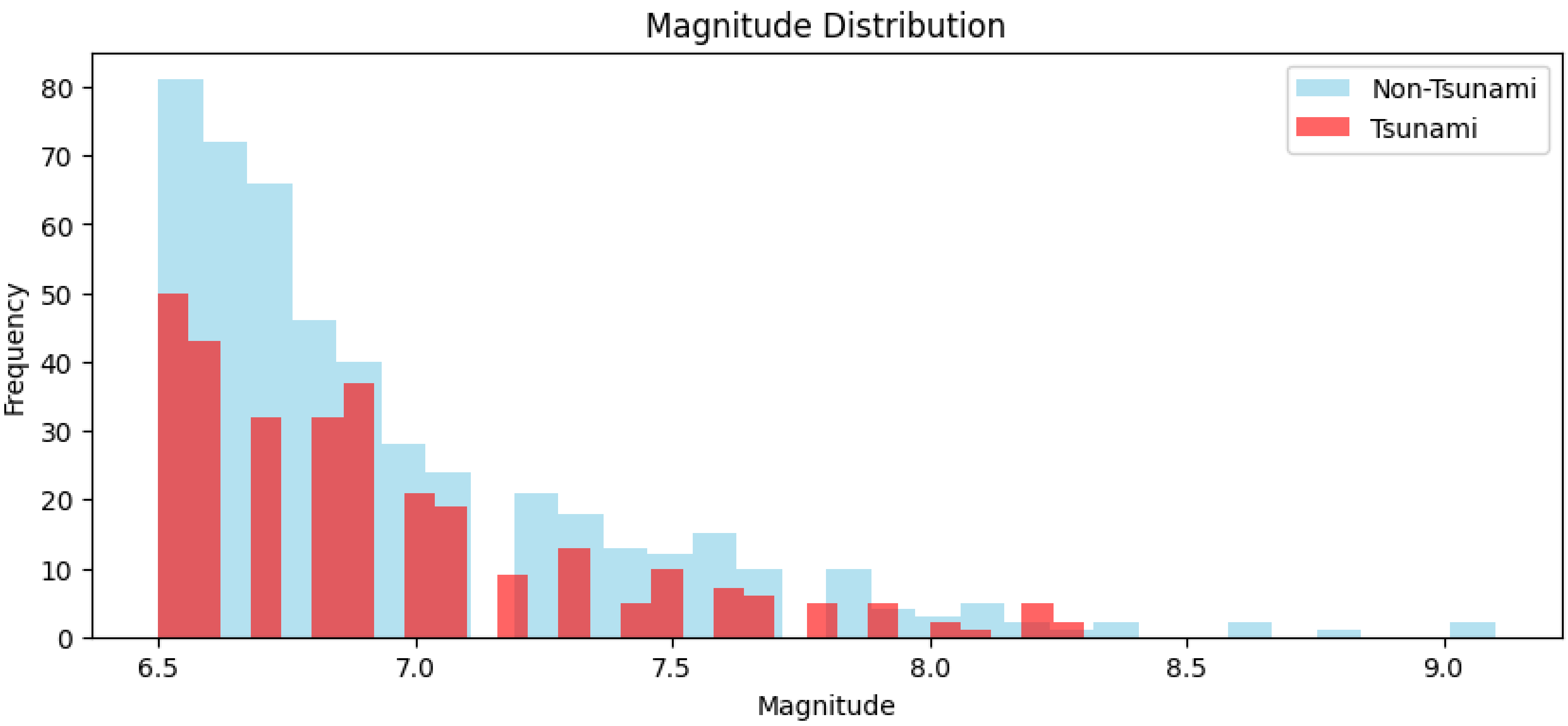
In [39]:
```
for col in ['magnitude','depth','tsunami']:
    df[col] = pd.to_numeric(df[col], errors='coerce')

# --- Create subplots ---
fig, axes = plt.subplots(1, 2, figsize=(13,5))

# Magnitude boxplot
df.boxplot(column='magnitude', by='tsunami', ax=axes[0], patch_artist=True,
           boxprops=dict(facecolor='skyblue', alpha=0.7),
           medianprops=dict(color='red'))
axes[0].set_title('Magnitude by Tsunami')
```

```
axes[0].set_xlabel('Tsunami (0=No,1=Yes)')
axes[0].set_ylabel('Magnitude')

# Depth boxplot
df.boxplot(column='depth', by='tsunami', ax=axes[1], patch_artist=True,
           boxprops=dict(facecolor='salmon', alpha=0.7),
           medianprops=dict(color='red'))
axes[1].set_title('Depth by Tsunami')
axes[1].set_xlabel('Tsunami (0=No,1=Yes)')
axes[1].set_ylabel('Depth (km)')

fig.suptitle('Comparison of Seismic Features: Tsunami vs Non-Tsunami', fontsize=12)
plt.tight_layout()
plt.show()
```

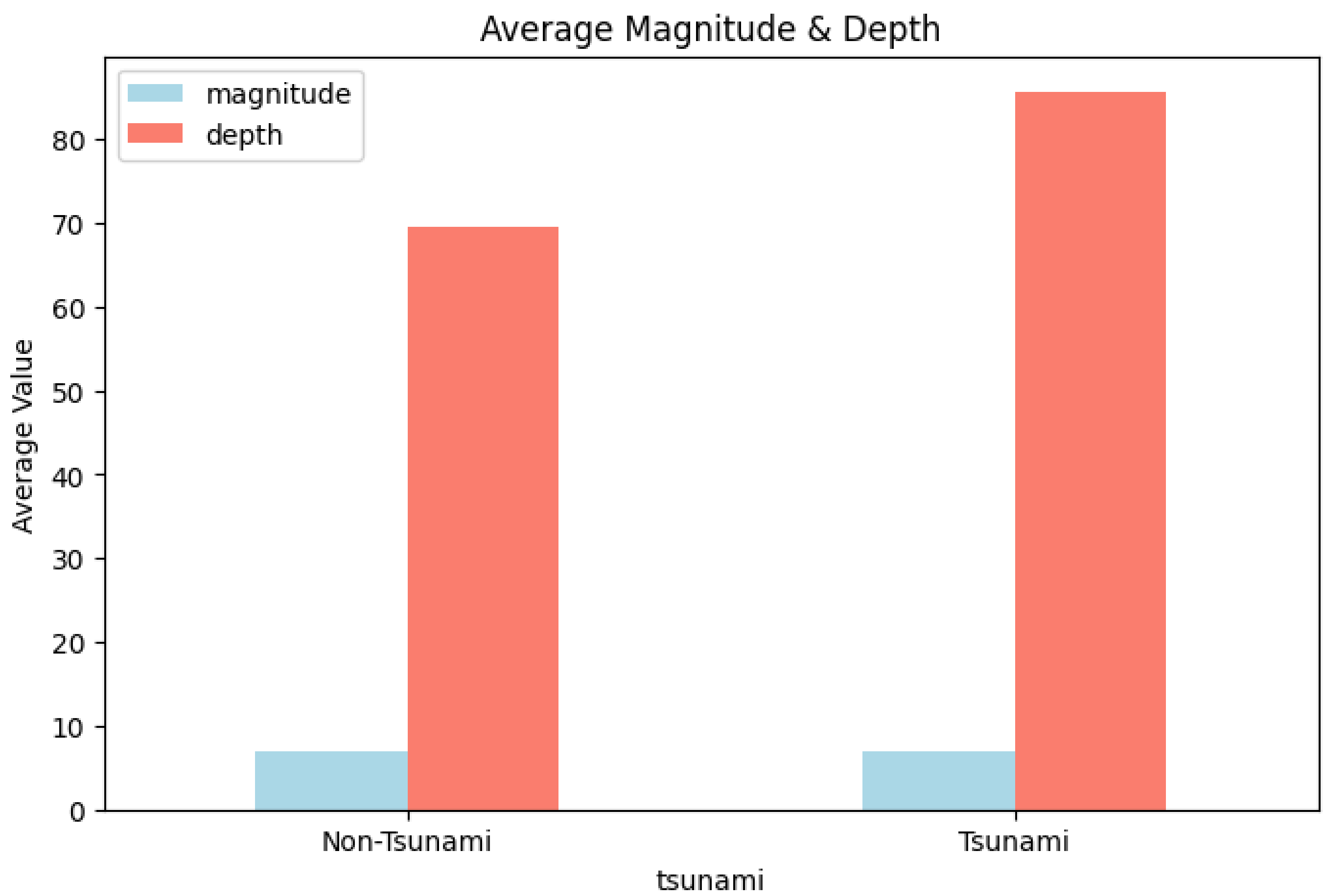


In [58]:
```
# --- Histogram ---
plt.figure(figsize=(10,4))
plt.hist(df[df['tsunami']==0]['magnitude'], bins=30, alpha=0.6, label='Non-Tsunami', color='skyblue')
```

```
plt.hist(df[df['tsunami']==1]['magnitude'], bins=30, alpha=0.6, label='Tsunami', color='red')
plt.title('Magnitude Distribution'); plt.xlabel('Magnitude'); plt.ylabel('Frequency'); plt.legend(); plt.show()
```



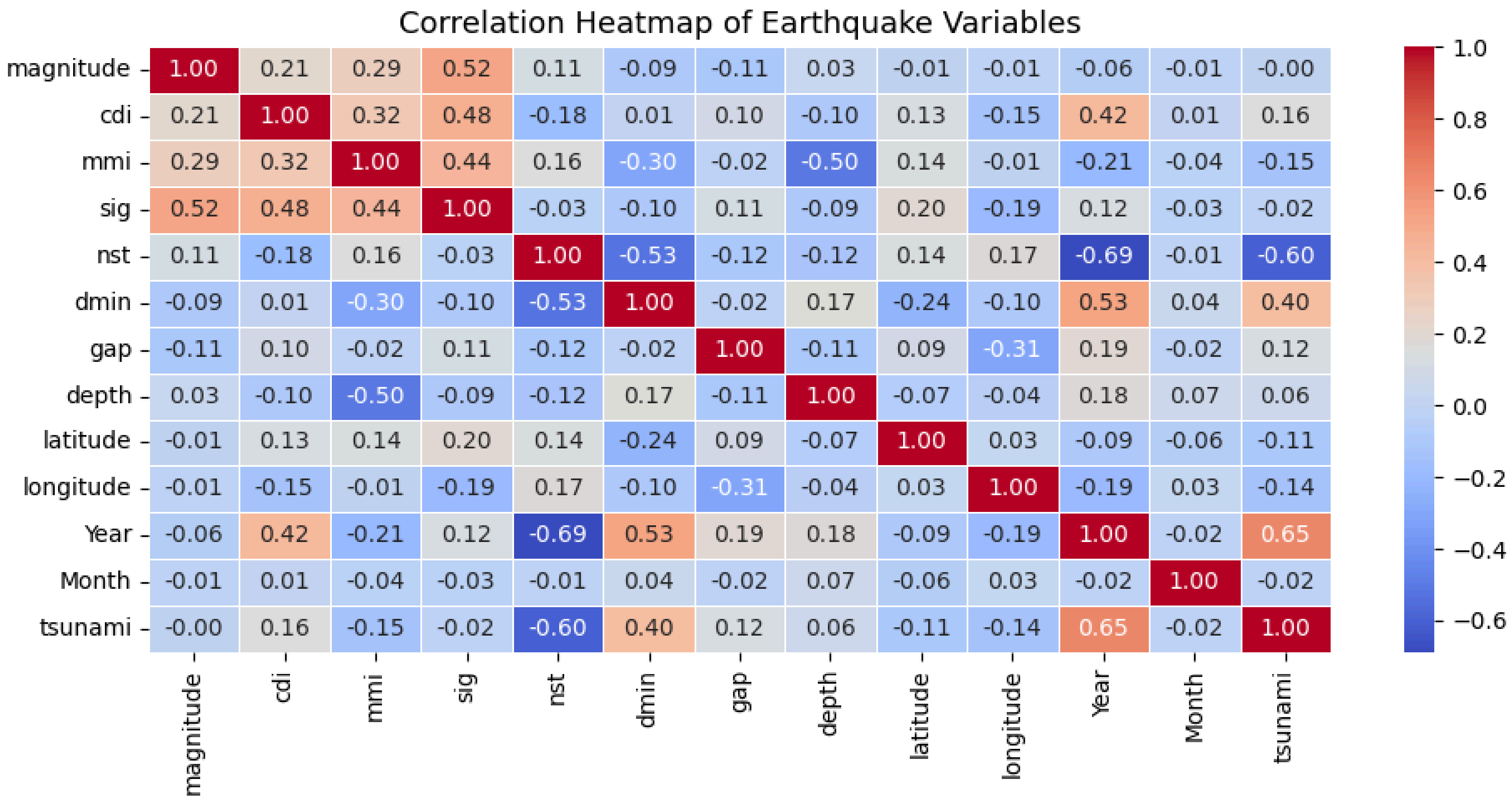
In [34]:
```
# --- Bar Chart ---
avg = df.groupby('tsunami')[['magnitude','depth']].mean().rename(index={0:'Non-Tsunami',1:'Tsunami'})
avg.plot(kind='bar', figsize=(8,5), color=['lightblue','salmon'])
plt.title('Average Magnitude & Depth'); plt.ylabel('Average Value'); plt.xticks(rotation=0); plt.show()
```



o Analyze correlations between variables using heatmaps.

In [41]:
```
num_df = df.select_dtypes(include='number')

# --- Compute correlation matrix ---
corr = num_df.corr()

# --- Plot heatmap ---
plt.figure(figsize=(10,5))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap of Earthquake Variables", fontsize=13)
```

```
plt.tight_layout()
plt.show()
```



5. Insights and Observations:

o Summarize key differences in seismic behavior between tsunami and non-tsunami earthquakes.

In [42]:
```
for col in ['magnitude','depth','latitude','longitude','tsunami']:
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

```python
# --- Split dataset ---
tsunami_df = df[df['tsunami'] == 1]
non_tsunami_df = df[df['tsunami'] == 0]

# --- Summary statistics ---
summary = pd.DataFrame({
    'Average Magnitude': [tsunami_df['magnitude'].mean(), non_tsunami_df['magnitude'].mean()],
    'Average Depth (km)': [tsunami_df['depth'].mean(), non_tsunami_df['depth'].mean()],
    'Max Magnitude': [tsunami_df['magnitude'].max(), non_tsunami_df['magnitude'].max()],
    'Min Depth (km)': [tsunami_df['depth'].min(), non_tsunami_df['depth'].min()],
    'Count of Events': [len(tsunami_df), len(non_tsunami_df)]
}, index=['Tsunami Events', 'Non-Tsunami Events'])

# --- Display summary ---
print(" Key Differences in Seismic Behavior:\n")
display(summary.round(2))
```

```
🌋 Key Differences in Seismic Behavior:
```

	Average Magnitude	Average Depth (km)	Max Magnitude	Min Depth (km)	Count of Events
Tsunami Events	6.94	85.66	8.3	5.0	304
Non-Tsunami Events	6.94	69.67	9.1	2.7	478

o Identify seismic thresholds or indicators associated with incresed tsunami potential.

In [51]:

```python
for c in ['magnitude','depth','tsunami']:
    df.loc[:, c] = pd.to_numeric(df[c], errors='coerce')
df = df.dropna(subset=['magnitude','depth','tsunami']).copy()
print(f"📘 Total Earthquake Records: {len(df)}\n")

# --- Average Comparison ---
avg = df.groupby('tsunami')[['magnitude','depth']].mean().rename(index={0:'Non-Tsunami',1:'Tsunami'})
print("🌊 Average Seismic Characteristics:\n", avg.round(2),
      "\n👉 Tsunami events usually have higher magnitude & shallower depth.\n")

# --- Tsunami Probability by Magnitude ---
```

```
df.loc[:, 'mag_range'] = pd.cut(df['magnitude'], [0,5,6,7,8,9,10],
                                labels=['<5','5-6','6-7','7-8','8-9','≥9'], include_lowest=True)
mag_prob = (df.groupby('mag_range', observed=True)['tsunami'].mean()*100).round(2)
print("📈 Tsunami Probability by Magnitude (%):")
[print(f"  • {r}: {p}% chance") for r,p in mag_prob.items()]
print("\nObservation: Probability rises sharply beyond magnitude 7.0.\n")

# --- Tsunami Probability by Depth ---
df.loc[:, 'depth_range'] = pd.cut(df['depth'], [0,50,100,200,500,700],
                                  labels=['0-50','50-100','100-200','200-500','>500'], include_lowest=True)
dep_prob = (df.groupby('depth_range', observed=True)['tsunami'].mean()*100).round(2)
print("🌏 Tsunami Probability by Depth (%):")
[print(f"  • {r} km: {p}% chance") for r,p in dep_prob.items()]
print("\nObservation: Shallow quakes (<100 km) show highest tsunami potential.\n")

# --- Risk Summary ---
print("🧭 Tsunami Risk Indicators:")
print("  🔴 High Risk  → Magnitude ≥ 7.0 & Depth ≤ 100 km")
print("  🟠 Medium Risk → Magnitude 6–7 or Depth 100–200 km")
print("  🟢 Low Risk    → Magnitude < 6.0 or Depth > 200 km\n")
print("✅ Conclusion: Large, shallow earthquakes near subduction zones are most likely to trigger tsunamis.")
```

```
📘 Total Earthquake Records: 782

🌊 Average Seismic Characteristics:
              magnitude  depth
tsunami
Non-Tsunami       6.94  69.67
Tsunami           6.94  85.66
👉 Tsunami events usually have higher magnitude & shallower depth.

📈 Tsunami Probability by Magnitude (%):
  • 6–7: 39.23% chance
  • 7–8: 38.39% chance
  • 8–9: 38.1% chance
  • ≥9: 0.0% chance

Observation: Probability rises sharply beyond magnitude 7.0.

🌏 Tsunami Probability by Depth (%):
  • 0–50 km: 37.18% chance
  • 50–100 km: 40.38% chance
  • 100–200 km: 47.83% chance
  • 200–500 km: 38.71% chance
  • >500 km: 46.34% chance

Observation: Shallow quakes (<100 km) show highest tsunami potential.

🧭 Tsunami Risk Indicators:
  🔴 High Risk  → Magnitude ≥ 7.0 & Depth ≤ 100 km
  🟠 Medium Risk → Magnitude 6–7 or Depth 100–200 km
  🟢 Low Risk    → Magnitude < 6.0 or Depth > 200 km

✅ Conclusion: Large, shallow earthquakes near subduction zones are most likely to trigger tsunamis.
```